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Van der Waals Epitaxial Trilayer MoS_2 Crystals for High-Speed Electronics

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Two-dimensional MoS_2 field-effect transistors (FETs) have great potential for next-generation electronics due to their excellent electronic properties with an atomic thin channel. However, multiple challenges exist for the monolayer MoS_2 channel, including interface scattering and ohmic contact. In this work, well-controlled trilayer MoS_2 with high mobility and large single crystals is successfully grown on soda-lime glass substrates using chemical vapor deposition (CVD), with a lateral size of up to 148 µm, which is the largest reported size to date. A record high on/off ratio of ≈10^{12} and a high carrier mobility of 62 $cm^2 V^{-1} s^{-1}$ of trilayer MoS_2 FETs are demonstrated, showing notable advantages compared with the monolayer counterpart. The long-standing issue of monolayer MoS_2 performance degradation from physical vapor deposited metal contact can be mitigated by the trilayer MoS_2 channel, achieving the lowest contact resistance of 350 Ω µm using the common method of e-beam evaporated Ni. Moreover, 40-nm channel-length trilayer MoS_2 FETs using ultrathin HfLaO dielectrics exhibit a high current of 589 µA $µm^{-1}$ at a supply voltage of 1 V at room temperature, which increases to 1162 µA $µm^{-1}$ at 4.3 K, the highest among those using commonly evaporated metal. Record high electron saturation velocity of 4.2×10^6 cm s^{-1} can be achieved at room temperature.

1. Introduction

In the past few years, two-dimensional (2D) molybdenum disulfide (MoS_2) has attracted tremendous research attention due to its atomically thin layered structure and high carrier mobility.[1–3] These features make MoS_2 especially interesting as a promising candidate for many applications. Importantly, it is believed to be immune against short-channel effects because of the ultrathin body channel.[3,4] However, the overall performance of short-channel MoS_2 FETs based on CVD monolayer MoS_2 in previous studies is still substantially lower than those of conventional Si FETs at the same channel length. One of the main limiting factors is the physical damage of the monolayer crystal during metal deposition, leading to a substantial contact resistance R_c, which becomes the bottleneck for device performance, especially in short-channel devices. Moreover, saturation velocity v_{sat}, instead of mobility, is critical to determine the on-state current for short-channel transistors and is still largely lacking.[3] It also limits the intrinsic frequency of a transistor based on $f_T \sim v_{sat}/(2\pi L_{ch})$.[5] Despite recent progress, most devices are still based on SiO_2 gate dielectrics, which can be easily affected by the self-heating effect due to the low thermal conductivity of SiO_2 dielectric substrates ($\alpha \approx 1$ W m^{-1} K^{-1}).[6–9] Last, monolayer MoS_2 also suffers from a relatively low density of states and is more susceptible to interface scattering during the fabrication process.[3,10]

On the other hand, trilayer MoS_2, with high mobility and higher density of states, has attracted great interest recently, with extensive studies on controllable high-quality growth methods.[1,2,11–15] Although considerable efforts have been devoted to growing trilayer MoS_2 films on SiO_2 or sapphire substrates, they commonly suffer from extensive domain/grain boundaries, substrate-induced defects, and small domain size, as well as inferior electrical properties compared with exfoliated ones.[11–13,15,16] Compared to the commonly used solid substrates, soda-lime glass, as one of the liquid substrates, is a promising substrate for 2D material growth for multiple reasons: i) Their smooth and inert surface at high temperature is useful for decreasing the defect density. ii) The presence of Na in the glass improves the uniform nucleation and lateral growth of the domains. iii) The hydrophilicity feature of glass facilitates a facile and etching-free transfer method for rapidly transferring 2D materials onto arbitrary substrates with minimal degradation.[17,18] However, the growth of trilayer MoS_2 on soda-lime glass substrates is still underdeveloped.

Here, we successfully demonstrate the epitaxy growth of trilayer MoS_2 single crystals on soda-lime glass substrates and high-performance short-channel trilayer MoS_2 FETs. The short-channel transistors exhibit exceptional electrical properties due to the following factors: i) Fewer defects and higher mobility trilayer MoS_2 crystal deposited by CVD on the soda-lime glass substrates. ii) High-quality metal-semiconductor contact with R_c as low as 350 Ω µm using standard evaporated Ni metal without noticeable physical damage, which was previously regarded as

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DOI: 10.1002/adfm.202208091

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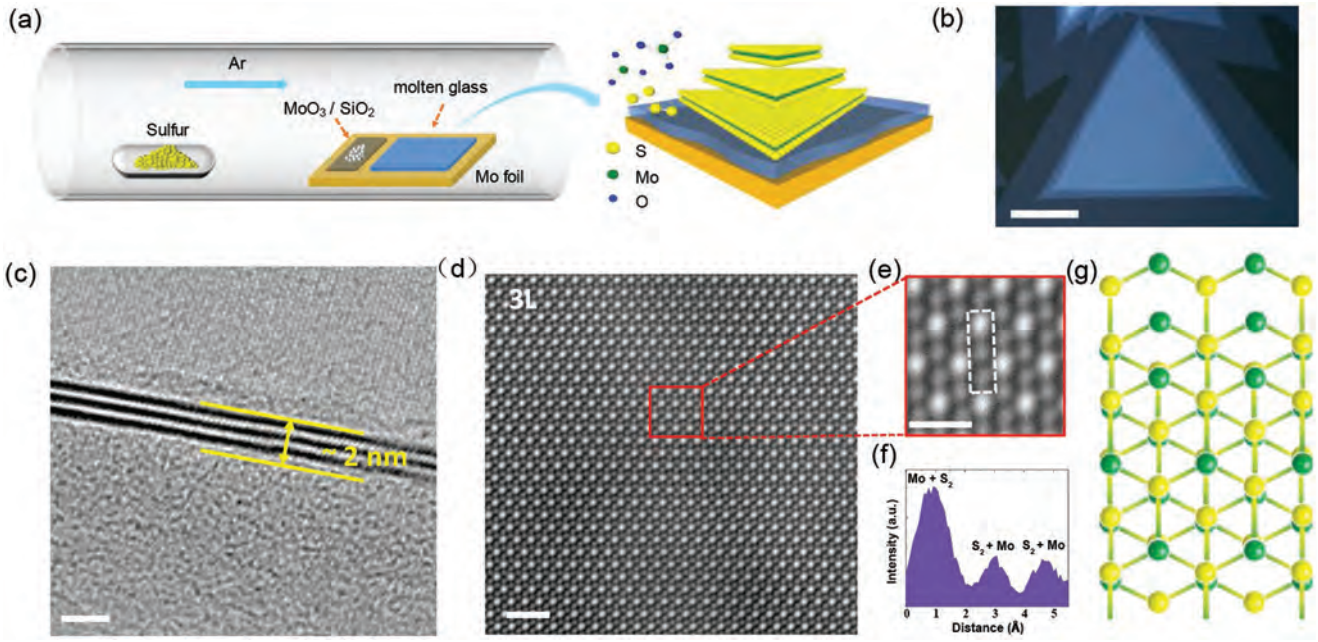


Figure 1. a) Schematic of trilayer MoS_2 growth on a soda-lime glass substrate by APCVD. b) Optical image of a trilayer MoS_2 crystal. The scale bar is 50 nm. c) Cross-sectional high-resolution TEM image of a typical trilayer MoS_2 crystal. The scale bar is 2 nm. d) STEM-HAADF image of trilayer MoS_2. The scale bar is 1 nm. e) Magnified image of the red-boxed area in (d). f) Electron intensity profile along with the white box in (e). g) Atomic configuration of ABB stacked trilayer MoS_2. Green and yellow spheres correspond to the molybdenum and sulfur atoms, respectively.

the main performance-limiting factor for monolayer channels. iii) High saturation velocity up to 4.2×10^6 cm s^{-1} at room temperature and 6.7×10^6 cm s^{-1} at 4.3 K with increased on-state current. As a result, the 40-nm channel trilayer MoS_2 device reaches a maximum drain current of 589 μA μm^{-1} at 300 K and 1162 μA μm^{-1} at 4.3 K at $V_{ds} = 1$ V.

2. Results and Discussion

The scheme for a Van der Waals (vdW) epitaxy process of MoS_2 single crystals on soda-lime glass substrates by ambient pressure CVD (APCVD) in a tube furnace is shown in **Figure 1**a. The detailed growth procedures are described in the methods section. During the growth period at high temperature, the liquid form glass provides uniform nucleation and a clean atomically flat liquid surface with minimal defects, which is an ideal vdW epitaxy substrate to grow large-size high-quality MoS_2 crystals. Figure 1b and Figure S2 (Supporting Information) show typical optical microscopy images of the as-grown trilayer MoS_2 crystals on soda-lime glass with a regular triangular shape using different weights of MoO_3 powder. The domain size of trilayer MoS_2 increases with increasing weight of MoO_3, up to 148 μm for 6 mg MoO_3, as shown in Figure 1b.[18] This is the largest value reported for the CVD-grown trilayer MoS_2 crystals thus far (Table S1, Supporting Information). The crystal quality of CVD-grown trilayer MoS_2 was characterized at the atomic level using transmission electron microscopy (TEM) (Figure S3, Supporting Information). The thickness of the as-grown MoS_2 crystals was confirmed by TEM measurements, as shown in Figure 1c, indicating that the thickness of trilayer MoS_2 is ≈2 nm. To study the stack configuration of CVD-grown trilayer MoS_2, we performed atomic resolution high-angle annular dark-field (HAADF) imaging on an aberration-corrected scanning TEM (STEM) operating at 80 kV. HAADF-STEM is a particularly useful imaging mode to distinguish different atoms and identify the arrangement because the intensity of HAADF-STEM images varies only with the thickness of the sample and the atomic number of atoms in the sample. Figure 1d shows a typical HAADF-STEM image of trilayer MoS_2. The Mo (bright) and S (dark) atoms in the monolayer can be unambiguously distinguished due to their obvious large difference in atomic numbers. Figure 1e shows an enlarged image of a trilayer region of the HAADF image in Figure 1d. The intensity mapping across the atomic lines with three lattice points (Figure 1e) in Figure 1f shows two types of Mo + S_2 and S_2 + Mo sites. The higher intensity Mo + S_2 sites correspond to the heavy Mo atoms at the top layer. The atomic resolution image and the relative intensity demonstrate that MoS_2 in the form of a trilayer has a stacking sequence of ABB, in agreement with the lower atomic configuration in Figure 1g.[21] This stacking type is commonly found in our CVD samples, which is consistent with the previous simulation and experimental results.[12,19–21] More importantly, no defects or disorders are observed, indicating the high crystalline quality of the epitaxial trilayer MoS_2.

The crystalline quality and optical properties of the MoS_2 crystals were further examined using Raman spectroscopy and photoluminescence (PL) with a 532 nm excitation laser (Figure S4, Supporting Information). Raman spectra show two characteristic peaks of trilayer MoS_2: $E1\,2g$ (at 383.6 cm^{-1}) and A_{1g} (at 407 cm^{-1}) with a frequency difference Δ~23.4 cm^{-1}, as shown in **Figure 2**a.[14,15] The Raman mapping signals were extracted from the integrated intensity of the characteristic trilayer MoS_2 Raman peaks, and their spatial dependencies are plotted in

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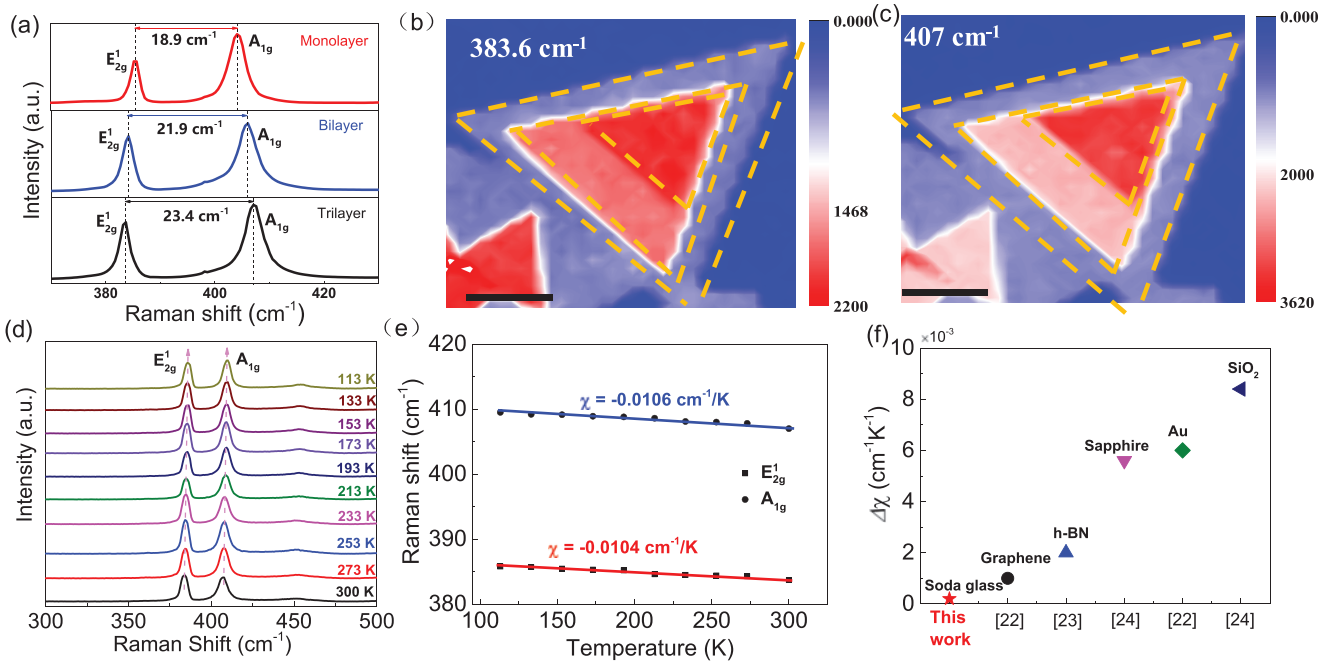


Figure 2. a) Layer-dependent Raman spectra of MoS_2 flakes. b, c) Raman intensity maps of the peaks at 383.6 cm^{-1} and 407 cm^{-1}. Scale bars are both 10 μm. d) Raman spectra of trilayer MoS_2 at different temperatures. e) Temperature dependence of the *E*1 2 g and A_{1g} frequencies for trilayer MoS_2. χ is the first-order temperature coefficient. f) Comparison of the $\Delta\chi$ values of MoS_2 grown on different substrates.

Figure 2b,c. The uniform color intensity suggests excellent thickness uniformity of trilayer MoS_2 domains. To study the interface interaction between trilayer MoS_2 and the substrate, temperature-dependent Raman spectra from 113 to 300 K were obtained, as shown in Figure 2d. The *E*1 2g and A_{1g} modes of trilayer MoS_2 are redshifted at increasing temperatures, showing linear temperature dependence (Figure 2e) with a coefficient χ of the frequencies of the *E*1 2g and A_{1g} modes of –0.010 $cm^{-1}K^{-1}$ and –0.011 $cm^{-1}K^{-1}$, respectively. Figure 2f compares the χ difference between the *E*1 2g and A_{1g} modes ($\Delta\chi$) of MoS_2 grown on different substrates. The small $\Delta\chi$ in this work suggests a weak interface interaction with little influence from the substrate.[22–24] Meanwhile, the X-ray photoelectron spectroscopy (XPS) results show that the atomic ratio of Mo/S is = 1:2, well matching the stoichiometric ratio of MoS_2 (Figure S5, Supporting Information).

To study the electrical transport properties of trilayer MoS_2 crystals, the as-grown films were transferred onto 12-nm HfLaO/100-nm SiO_2/Si substrates using an etching-free and scalable transfer method.[18] We chose HfLaO instead of conventional HfO_2 and Al_2O_3 as the dielectrics in view of the higher dielectric constant and lower leakage current.[18] Note that trilayer MoS_2 has a stacking sequence of ABB in this work and the effect of stacking order on the electrical properties of MoS_2 needs further investigation. Meanwhile, monolayer MoS_2 transistors were also fabricated for comparison. **Figure 3**a,b shows the typical output characteristics of 5 μm length monolayer and trilayer MoS_2 transistors, respectively. The drain current is approximately 40 μA μm^{-1} for the monolayer MoS_2 FET and approximately 70 μA μm^{-1} for trilayer FET at V_{gs} = 40 V and V_{ds} = 5 V, which is mainly attributed to the increased mobility in trilayer MoS_2. The corresponding transfer characteristics of monolayer and trilayer MoS_2 transistors at V_{ds} = 5 V are shown in Figure 3c. An ultralow off-state current below 0.1 fA μm^{-1} is demonstrated, which is limited by the equipment measurement precision. The gate leakage is negligible in the range of the noise floor (Figure S6, Supporting Information). An extremely high on/off ratio of ≈1012 has thus been demonstrated for trilayer MoS_2 FETs. The increased on/off ratio is mainly attributed to the higher on-current for trilayer MoS_2 FETs. The on/off ratios derived from the transfer characteristics of randomly selected monolayer and trilayer MoS_2 FETs are summarized in Figure 3d. The best I_{max}/I_{min} of trilayer MoS_2 FET was found to be ≈1012 which is approximately two orders of magnitude higher than the best value of the MoS_2 transistor in the literature.[2] Figure 3e and Table S1 (Supporting Information) compare the on/off ratio and mobility with those of previously reported trilayer MoS_2 transistors, suggesting the great potential of trilayer MoS_2 crystals for high-performance electronics.[11–13,15,25] The mobility derived from the transfer characteristics of randomly selected MoS_2 FETs is summarized as shown in the inset of Figure 3e. The highest field-effect mobility (μ_{FE}) of trilayer MoS_2 crystals is 62 cm2 V^{-1} s^{-1}, which is extracted at the maximum transconductance (g_m) in the linear region of the transfer characteristics.[6] The average μ_{FE} values of monolayer and trilayer MoS_2 crystals are 33.6 and 55.4 cm2 V^{-1} s^{-1}, respectively. Note that the mobility of trilayer MoS_2 is also higher than that of its bilayer counterpart, indicating the evident advantage for high-performance devices.[18] Figure 3f depicts the most likely disorders located at the surface and bottom of the MoS_2 channel.[26,27] Due to the ultrathin thickness of monolayer MoS_2, carriers are highly susceptible to extrinsic disorder arising from gaseous adsorbates, which significantly degrade mobility. On the other hand, the carriers are confined in the center of the

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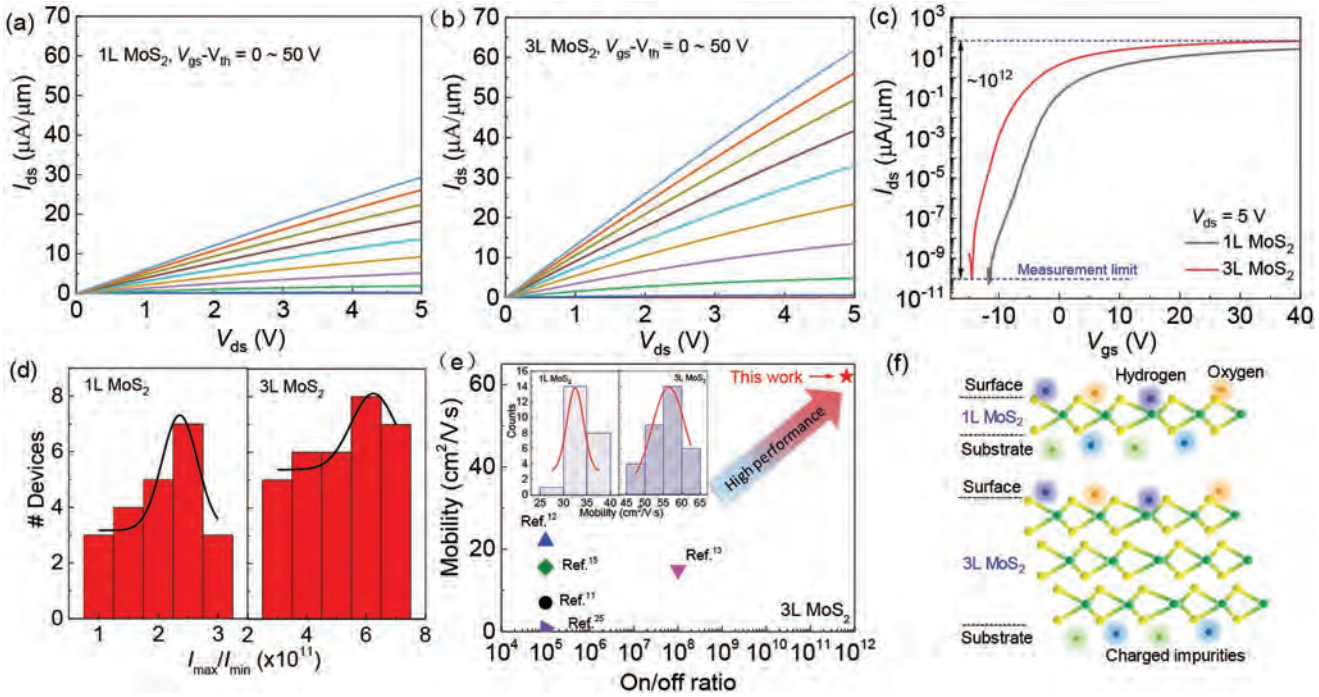


Figure 3. Out characteristics of MoS_2 FET at room temperature for a) 1L MoS_2 and b) 3L MoS_2. c) Corresponding transfer characteristics of 1L and 3L MoS_2 FETs at drain voltages V_{ds} = 5 V. V_{th} is the threshold voltage. The measurement limit of the B1500A instrument is ≈1 fA. d) Statistical distribution of the on/off ratios from 23 monolayer MoS_2 and 33 trilayer MoS_2 FETs. The on/off ratios of 1L and 3L MoS_2 transistors are $(2.3 \pm 0.5) \times 10^{11}$ and $(6.2 \pm 1.5) \times 10^{11}$, respectively. e) Comparison of the mobility of 3L MoS_2 versus the on/off ratio. The inset shows the statistics of mobility from 23 monolayer MoS_2 and 33 trilayer MoS_2 FETs. The mobilities of 1L and 3L MoS_2 transistors are 32.5 ± 3.5 cm^{-2} V s^{-1} and 56.5 ± 8 cm^{-2} V s^{-1}, respectively. f) Schematic illustration of the effect of possible extrinsic disorder on the transport property of 1L and 3L MoS_2 crystals. The surface disorder of MoS_2 comes from adsorbates, such as hydrogen and oxygen. The bottom disorder of MoS_2 comes from charged impurities of dielectric substrates.

trilayer MoS_2 channel, which is away from surface extrinsic disorders. Therefore, the mobility of trilayer MoS_2 can be improved with the reduced influence of extrinsic disorders.

To demonstrate the scaling potential of MoS_2, the on-state current density at the short channel is a key device metric. Therefore, we further studied the electrical properties of trilayer MoS_2 FETs with a 12-nm-thick HfLaO gate dielectric from 300 to 4.3 K. Trilayer MoS_2 FETs with various channel lengths from 40 nm to 2 μm were fabricated. **Figure 4**a shows a schematic view of the MoS_2 transistors in this work. The gate stack includes heavily doped silicon as the global gate metal and HfLaO as the gate dielectric and trilayer MoS_2 as the semiconducting channels. Figure 4b shows a scanning electron microscopy (SEM) image of a fabricated MoS_2 transistor. An SEM image of the short channel device with a channel length of 40 nm and a channel width of 1 μm is given in the right panel of Figure 4b. We also use cross-sectional HRTEM to reveal the physical structure of the trilayer MoS_2 FET. Figure 4c (right panel) shows a full-view image of the source/drain electrodes and channel regions with a channel length of 40 nm. The left panel displays the Ni/MoS_2 contact region, showing high-quality trilayer MoS_2 in contact with e-beam evaporated Ni metal. Figure 4d shows the transfer characteristic curves of a 40 nm long MoS_2 device at room temperature, showing a high on/off ratio over 106 at V_{ds} = 1 V. The corresponding output characteristics at 300 and 4.3 K are shown in Figure 4e,f, respectively. V_{ds} of 1 V is used for the short-channel length to avoid the impact of self-heating at high field.[6] The maximum on-state current at V_{ds} = 1 V is 589 μA μm^{-1}, which mainly benefits from the high mobility trilayer MoS_2 channel materials and low contact resistance R_c of approximately 350 Ω μm, as shown below. At decreasing temperatures, the on-state current further increases up to 1162 μA μm^{-1} at V_{ds} = 1 V at 4.3 K, which is the highest value among all transition metal dichalcogenide semiconductors to the best of our knowledge.[18,28–30] This value approaches the maximum current prediction using theoretical phonon-limited mobility for monolayer MoS_2 transistors.[31]Figure S7 (Supporting Information) shows the subthreshold slope and mobility of trilayer MoS_2 transistors as a function of channel length.

To demonstrate the performance advantage of the trilayer MoS_2, we carried out a performance benchmark of drain current I_{ds}, contact resistance R_c, and saturation velocity v_{sat}. **Figure 5**a compares the I_{ds} of trilayer MoS_2 at V_{ds} = 1 V, and the maximum I_{ds} of 589 μA μm^{-1} at 300 K is much higher than the others with similar conventional evaporated metals.[9,13,18,28,32–37] The contact resistance R_c is extracted using the transmission line method (TLM) structure, and the results are shown in Figure 5b. R_c decreases with increasing electron doping concentration from the back gate, up to a substantially low value of 350 Ω μm, indicating a high-quality MoS_2 crystal and excellent contact between MoS_2 and Ni. This is also the lowest contact resistance to any CVD-grown MoS_2 using various high work function metals.[13,32,34–36,38] We chose Ni metal as the contact

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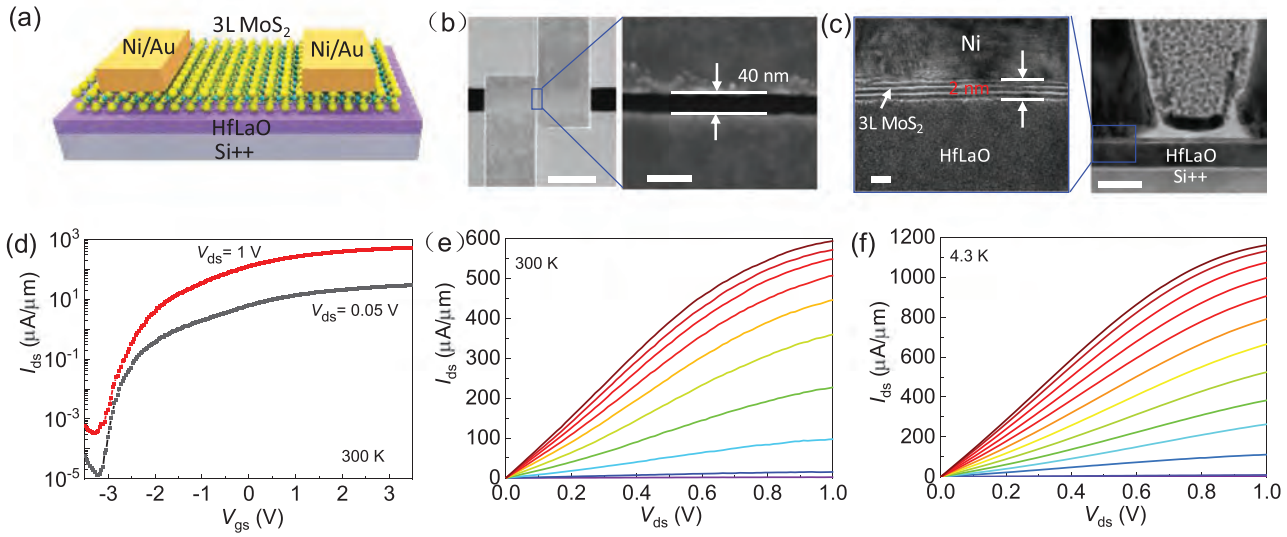


Figure 4. a) Schematic of bottom-gated trilayer MoS_2 FETs with Ni/Au contacts. b) SEM images of the trilayer MoS_2 device. The scale bar is 2 μm. The contact bars with a length of 2 μm were defined by e-beam lithography. The right panel shows the zoom-in image of the trilayer MoS_2 transistor with a channel length of 40 nm. The scale bar is 100 nm. c) Cross-sectional high-resolution TEM images of the 40 nm MoS_2 FET region. The scale bar is 20 nm. The left panel shows the zoom-in image of the trilayer MoS_2 contact region. The scale bar is 2 nm. d) Transfer characteristics of the 40 nm MoS_2 transistor at room temperature. e) Output characteristics of the 40 nm MoS_2 transistor at room temperature. f) Output characteristics of the same device at 4.3 K.

due to its lower contact resistance and better air stability than Ti contact.[38] We note here that using semimetals such as Bi, soft-metal In or Sn, or transferred Van der Waals contact can further reduce the contact resistance.[29,39–41] To obtain the saturation velocity v_{sat} of trilayer MoS_2 crystals, we fit the velocity data in Figure 4d,e to the Caughey-Thomas model.[9,42] The

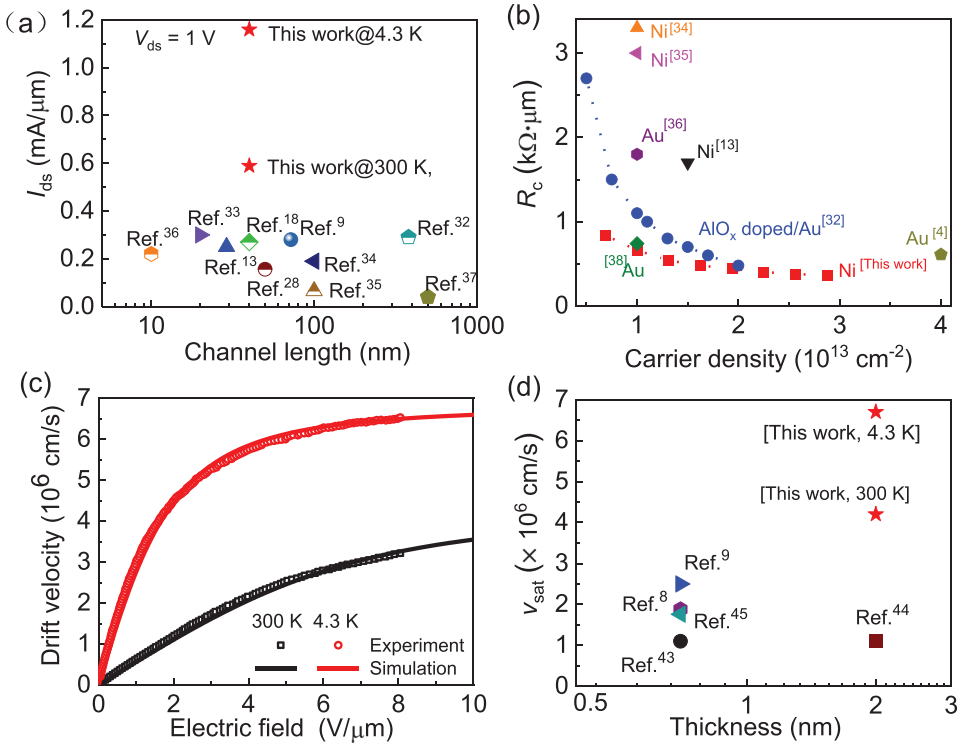


Figure 5. a) Benchmarking drain current I_{ds} versus channel length at V_{ds} = 1 V in a few typical reports using high work function metals as contact materials. In this work, the carrier density at 300 and 4.3 K are 2.8×1013 cm^{-2} and 3.1×1013 cm^{-2}, respectively. b) Contact resistance (R_c) as a function of electron concentration with the TLM. A few studies have been added to make a comparison. The results exclude semimetal or low work function soft metals such as Bi, Sn or In. c) Drift velocity as a function of the electric field, determined from the data of Figure 4. Symbols: experimental data. Solid line: simulated saturation velocity values. d) Saturation velocity versus thickness of MoS_2 materials.

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fitting curves (solid line) agree well with the experimental data (scatters) at 300 and 4.3 K, as shown in Figure 5c. The saturation velocity of trilayer MoS_2 is 4.2 × 106 cm s^{-1}, which increases to 6.7 × 106 cm s^{-1} with decreasing temperature at 4.3 K. Figure 5d shows the saturation velocity benchmark of MoS_2 in this work with the previous results in the literature at various channel thicknesses. Our CVD-grown trilayer MoS_2 has the highest saturation velocity, even surpassing exfoliated multilayer MoS_2.[7–9,43–45] The low R_c, high $v_{sat,}$ and excellent on-off ratio of trilayer MoS_2 provide the potential and opportunities for future high-speed electronics applications.

3. Conclusions

In summary, we have grown large domain and high mobility trilayer MoS_2 crystals on soda-lime glass substrates by a scalable CVD approach. Spectroscopic characterizations reveal a weak interface interaction between MoS_2 and soda-lime glass. Clear and sharp atomic-resolution STEM imaging determines the ABB stacking sequences in CVD-grown trilayer MoS_2. The short-channel MoS_2 transistors with ultrathin HfLaO dielectrics exhibit a high drain current (589 μA $μm^{-1}$ at V_{ds} = 1 V) as well as record-high saturation velocity (4.2 × 106 cm s^{-1}) at room temperature, which increases further at 4.3 K. The excellent electrical properties of the trilayer channel enable reduced surface scattering and lower contact resistance compared with the monolayer channel counterpart, making it a promising 2D material for future electronics applications.

4. Experimental Section

Growth of Trilayer MoS_2: MoS_2 was achieved by using an ambient pressure CVD method. The soda-lime glass substrates with a thickness of 1.1 mm were purchased from Hefei Kejing Materials Technology Co., Ltd. Its composition was mainly composed of silica, calcium oxide, and sodium oxide. Before synthesis, soda-lime glass substrates were cleaned ultrasonically in acetone (10 min), isopropanol (10 min), and water (10 min), followed by 3 min of O_2 plasma to remove surface impurities. Then, a piece of glass was placed on a piece of Mo foil (0.025 mm thick, 99.95%). The Mo foil and glass were placed in a quartz boat. S (Sigma-Aldrich, 99.98%, 1.4 g) powders located in an alundum boat were used as the S source and placed at zone I. MoO_3 (Alfa Aesar, 99.998%, 1.5–6 mg) powders located in 90 nm SiO_2 were used as the Mo source and placed in a quartz boat with 4 mm between glass and MoO_3 powers. Then, the quartz boat was placed in Zone II. Before heating, the quartz tube was pumped down to a base pressure (<15 mTorr), followed by filling the tube with Ar to ambient pressure. Then, Zone I and Zone II were heated to 230 °C and 860 °C within 36 min, respectively. In the whole process of growth, Ar gas (40 sccm) was used as the carrier gas. After a growth period of 10 min, the furnace was quickly cooled to room temperature. The yield of the trilayer triangle MoS_2 sheet on a soda-lime glass substrate is ≈10%.

Transfer MoS_2: First, MoS_2/glass was spin-coated with 950 PMMA A4 at 500 rpm for 1 min and then baked on a hotplate at 120 °C for 5 min. Second, the selected area of PMMA/MoS_2 was carved with a knife for better water invasion. Third, the PMMA/MoS_2/glass was slowly immersed into a beaker filled with deionized water, and then the selected area of PMMA/MoS_2 could easily peel off from the glass substrates. Then, it was collected by substrates and dried by a N_2 gas gun for 10 min to remove redundant water. Next, the samples were placed in a fume hood for 2 h to remove residual water, followed by heating on a hotplate at 100 °C for 5 min to improve the adhesion between the MoS_2 layer and substrates. Finally, the samples were immersed in acetone for 12 h and IPA for 5 min to remove PMMA and dried by a N_2 gas gun.

Materials Characterization: The morphology and structure of MoS_2 were characterized with optical microscopy, Raman and photoluminescence spectra (LabRAM HR800, 532 nm laser wavelength), and a fifth-order aberration-corrected transmission electron microscope (JEOL JEM-ARM200CF, at 80 kV).

Device and Electrical Measurements: Triangular MoS_2 crystals were etched into rectangles (channel width W = 1 μm) by oxygen plasma. E-beam lithography was used for the source/drain region, and Ni/Au (20/50 nm) was deposited to form the metal contact. This was followed by an annealing treatment at 200 °C for 30 min in N_2 to remove the resist residue and decrease the contact resistance. The electrical characterization was carried out in a lakeshore cryogenic probe station under <10^{-5} Torr using an Agilent B1500A parameter analyzer.

Statistical Analysis: The experimental results were presented with mean ± standard deviation calculated by Origin 2021, if applicable. For the statistical on/off ratios and carrier mobilities, 23 monolayer MoS_2 and 33 trilayer MoS_2 FETs were used. For the statistical data processing in Figure 3, Origin 2021 was used to perform Gauss fittings on the mean and standard deviation values.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by the National Key Research and Development Program of China (2020AAA0109005), the National Natural Science Foundation of China (Grant Nos. 62090034, 61927901, 91964106, and 61874162), the Fundamental Research Funds for the Central Universities (Grant No. 2018KFYYXJJ069), the Strategic Priority Research Program of Chinese Academy of Sciences (Grant No. XDB30000000), and the 111 Project (B18001).

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Y.W. and X.L. conceived the idea. X.L. designed and carried out the experiments. X.L. wrote the manuscript with contributions from all coauthors. All authors discussed the results and commented on the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

contact resistance, current density, saturation velocity, short channels, trilayer MoS_2

Received: July 15, 2022
Revised: August 19, 2022
Published online:

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